

12.476

EE25BTECH11047 - RAVULA SHASHANK REDDY

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Question:

If

$$\mathbf{AB} = \mathbf{I},$$

then:

- a) $\mathbf{B} = \mathbf{A}^\top$ b) $\mathbf{A} = \mathbf{B}^\top$ c) $\mathbf{B} = \mathbf{A}^{-1}$ d) $\mathbf{B} = \mathbf{A}$

Solution:

Given:

$$\mathbf{AB} = \mathbf{I} \tag{1}$$

Multiply both sides by $\mathbf{A}^{-1}$:

$$\mathbf{A}^{-1}(\mathbf{AB}) = \mathbf{A}^{-1}\mathbf{I} \tag{2}$$

$$\Rightarrow (\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{A}^{-1} \tag{3}$$

$$\Rightarrow \mathbf{IB} = \mathbf{A}^{-1} \tag{4}$$

$$\Rightarrow \mathbf{B} = \mathbf{A}^{-1} \tag{5}$$

$$\boxed{\mathbf{B} = \mathbf{A}^{-1}}. \tag{6}$$